BanditExplain: Scalable Probabilistic Explanations with PAC Guarantees via Multi-Armed Bandits

SifEddine Sellami
Université Sorbonne Paris Nord, LIPN

Louenas Bounia
Université Sorbonne Paris Nord, LIPN

Juba Agoun
Université Lumière Lyon 2, ERIC

Céline Rouveirol
Université Sorbonne Paris Nord, LIPN

Abstract

Formal explainability seeks to provide machine learning explanations with certified guarantees. Abductive explanations guarantee a prediction using a subset of features, but they can be too large to remain understandable. Probabilistic explanations relax this requirement, yielding more compact explanations, but computing them remains challenging. Existing approaches face a trade-off between formal guarantees and scalability: exact methods provide strong guarantees but do not scale to complex models, while scalable methods lack formal approximation guarantees. We bridge this gap by connecting probabilistic explainability to multi-armed bandits. Our approach, BanditExplain, frames greedy explanation as a sequence of best-arm identification problems, where model queries estimate the probability that candidate features preserve the prediction. This avoids exact model counting while providing PAC guarantees. BanditExplain is model-agnostic and requires only a polynomial number of model queries. Experiments on 16 benchmark datasets show that it outperforms existing scalable approaches on decision trees while closely matching exact greedy descent. It also scales to tree ensembles and neural networks, where exact counting is out of reach.

1 INTRODUCTION

Machine learning models are now routinely deployed in high-stakes domains, where predictive accuracy alone is no longer sufficient: stakeholders ask for clear and well-justified explanations (Barredo Arrieta et al., 2020). Popular model-agnostic methods such as LIME (Ribeiro et al., 2016), SHAP (Lundberg and Lee, 2017), and Anchors (Ribeiro et al., 2018) explain individual predictions by probing the model around the instance of interest. Because they are heuristic, they offer no formal guarantee that the explanation faithfully reflects the model's behavior (Huang and Marques-Silva, 2024), and they can be easily manipulated (Slack et al., 2020). These shortcomings have motivated *formal explainability*, which seeks explanations with certified guarantees (Darwiche and Hirth, 2020; Marques-Silva and Ignatiev, 2022).

A central question in XAI is to explain why a classifier h assigns a label $h(x)$ to an instance x described by d features. Feature-based explanations identify a subset $S \subseteq [d] = \{1, \ldots, d\}$ of features, where, whose values x_S support the prediction regardless of the other features. When fixing x_S determines $h(x)$ with certainty, S is called an *abductive explanation* (Ignatiev et al., 2019) or a *sufficient reason* (Darwiche and Hirth, 2020). Despite their strong foundations, abductive explanations are often too large to be useful in practice (Wäldchen et al., 2021). This limitation is rooted in human cognition: working memory holds only a few elements, classically estimated at *seven* plus or minus two (Miller, 1956) and recently at about *four* (Cowan, 2001), and user studies find that concise explanations are easier to understand (Narayanan et al., 2018).

This has motivated *probabilistic explanations*, which trade a small, quantified error for substantially more compact explanations (Wäldchen et al., 2021; Blanc et al., 2021; Arenas et al., 2022; Izza et al., 2023). Given a distribution $\mathcal{D}$ over instances and a size limit k, the objective is to find a subset S of at most k features that minimizes the probability that an instance agreeing with x on S receives a different label:

$$\epsilon_{h,x}(S) = \Pr_{z \sim \mathcal{D}} \left[h(z) \neq h(x) \;\middle|\; z_S = x_S \right]. \tag{1}$$

Following the formal XAI literature (Wäldchen et al., 2021; Arenas et al., 2022; Bounia and Koriche, 2023; Izza et al., 2024), we consider Boolean features and the uniform distribution over $\{0,1\}^d$, for which the error function $\epsilon_{h,x} : 2^{[d]} \to \mathbb{R}$ becomes

$$\epsilon_{h,x}(S) = \frac{|\{z : h(z) \neq h(x),\, z_S = x_S\}|}{|\{z : z_S = x_S\}|} = \frac{\mu_{h,x}(S)}{2^{d-|S|}}, \quad (2)$$

where z ranges over $\{0,1\}^d$ and $\mu_{h,x}(S)$ is the number of completions of x_S misclassified by h, a *model counting (MC)* quantity. For example, for a patient x predicted as diabetic, an abductive explanation may need to fix many features only to rule out a handful of rare configurations, whereas $S = \{Glucose, BMI\}$ with $\epsilon_{h,x}(S) = 0.02$ preserves the prediction for 98% of the possible patients. This gain in interpretability comes at a steep computational price: evaluating $\epsilon_{h,x}(S)$ for a single subset already requires model counting (MC). Under the uniform distribution, deciding whether an explanation of size k with error at most τ exists is $\mathsf{NP}^{\mathsf{PP}}$-hard in general (Wäldchen et al., 2021), and remains NP-hard even for decision trees (Arenas et al., 2022), for which abductive explanations can be computed in polynomial time (Izza et al., 2022). Worse, minimizing the size of such explanations is hard to approximate (Kozachinskiy, 2023). This raises the question of whether compact probabilistic explanations can be identified without repeatedly solving such costly counting problems.

Experiments on 16 benchmark datasets show that, on decision trees, where errors can be computed exactly, BanditExplain produces more accurate explanations than Anchors and locally-minimal explanations, while closely matching exact greedy descent. It also applies, with the same guarantees, to tree ensembles and neural networks, where exact counting is out of reach.

Contributions. In summary, this paper provides:

(i) **A bridge between formal explainability and multi-armed bandits:** the greedy computation of probabilistic explanations reduces to a sequence of best-arm identification problems. (ii) **A negative result on approximate model counting:** reproducing the decisions of exact greedy descent requires both a logical encoding of the classifier and an exponentially small approximation error. (iii) **BanditExplain,** a model-agnostic algorithm with PAC guarantees that provably recovers exact counting-based explanations under a separation condition and uses a polynomial number of model queries. (iv) **An empirical evaluation** showing that it outperforms scalable state-of-the-art approaches, closely matches exact greedy descent on decision trees, and scales to tree ensembles and neural networks.

2 RELATED WORK

Probabilistic Explanations and Their Computation. We say that S is a *k-size, τ-explanation* of x if $|S| \leq k$ and $\epsilon_{h,x}(S) \leq \tau$; abductive explanations correspond to $\tau = 0$. Finding such explanations is Σ_2^{P}-hard for random forests (Audemard et al., 2022) and neural networks (Barceló et al., 2020), and $\mathsf{W}[2]$-hard in k (Ordyniak et al., 2023); the only positive approximation result concerns linear models (Subercaseaux et al., 2025). Our setting is local and uses the uniform distribution over Boolean features; other works consider data distributions represented by probabilistic circuits (Wang et al., 2021), global explanations (Bassan et al., 2024), or query-based algorithms with provable precision guarantees (Blanc et al., 2021). Methods with formal guarantees rely on exact MC: exact solvers compute optimal explanations (Arenas et al., 2022; Izza et al., 2023) but do not scale beyond small models, and the greedy descent of Bounia and Koriche (2023), whose approximation guarantee stems from the supermodularity of $\mu_{h,x}$, requires $\mu_{h,x}$ to be computable in polynomial time, as for decision trees. Scalable methods, by contrast, come without optimality guarantees: locally-minimal explanations (Izza et al., 2024) use Monte Carlo sampling or approximate counting (Chakraborty et al., 2016) to ensure that no single feature can be removed, but give no bound on the distance to an optimal explanation, and sampling heuristics such as Anchors (Ribeiro et al., 2018) search for high-precision rules without approximation guarantees (Marques-Silva, 2023; Wu et al., 2023). Existing approaches thus face a trade-off between scalability and guarantees.

Bandit-based Explanation Search. Closest to our approach, Anchors (Ribeiro et al., 2018) uses the KL-LUCB algorithm (Kaufmann and Kalyanakrishnan, 2013) to select, at each step of a beam search, the candidate rules with the highest estimated precision. Both Anchors and BanditExplain thus rely on best-arm identification, but Anchors seeks rules meeting a precision threshold, without any guarantee on the optimality of the returned rule, whereas BanditExplain targets the error-minimizing candidate at each step and thereby recovers the approximation guarantees of counting-based greedy methods whenever suboptimal candidates are well separated. Best-arm identification itself has been studied in both the fixed-confidence (Even-Dar et al., 2006; Mannor and Tsitsiklis, 2004; Kalyanakrishnan et al., 2012) and fixed-budget (Audibert et al., 2010) settings; we build on the PAC framework of Even-Dar et al. (2006), and adap-

tive algorithms such as LUCB (Kalyanakrishnan et al., 2012) can be plugged into BanditExplain to further reduce the number of model queries. An extended discussion of related work is given in Appendix **??**.

3 BEST-ARM IDENTIFICATION

Most methods for computing probabilistic explanations with formal guarantees rely on model counting (MC) (Arenas et al., 2022; Izza et al., 2023; Bounia and Koriche, 2023; Izza et al., 2024; Subercaseaux et al., 2025), which is limited by the exponential size of the input space over free features. MC is #P-complete in general (Valiant, 1979), and even constant-factor approximation is NP-hard (Roth, 1996). Exact counting is tractable only for restricted representations, such as decision trees and d-DNNF circuits (Darwiche and Marquis, 2002), while compiling richer models may incur exponential blow-up (Shi et al., 2020). Approximate counters mitigate this bottleneck but still require costly repeated queries (Chakraborty et al., 2016; Gomes et al., 2009).

Yet a greedy algorithm does not use these counts for their own sake: like greedy procedures for submodular and supermodular optimization (Nemhauser et al., 1978; Bounia and Koriche, 2023), it only compares them, in order to decide which candidate is best. In this section, we introduce *best-arm identification* (Even-Dar et al., 2006; Audibert et al., 2010; Lattimore and Szepesvári, 2020), the tool that performs such comparisons directly. Its key property, which underlies our whole approach, is that the cost of identifying the best option depends on the separation between candidates *i.e.,* how different the options are, not on how large the space they are defined over is. We adapt a standard formulation to our setting, considering arms with Bernoulli rewards and a minimization objective, since the quantities of interest are error probabilities.

Setting. We are given n arms, indexed by $[n]$. Each arm i is associated with an unknown distribution over $\{0,1\}$ with mean θ_i, and *pulling* arm i returns an independent sample from this distribution. Let $\theta^* = \min_{i\in[n]} \theta_i$. A *best arm* is an arm i such that $\theta_i = \theta^*$, and the goal is to identify a best arm using as few pulls as possible. Given a precision $\eta > 0$, an arm i is *η-optimal* if $\theta_i \le \theta^* + \eta$. We also denote by

$$\Delta = \min\left\{\theta_i - \theta^* \,:\, i \in [n],\ \theta_i > \theta^*\right\} \tag{3}$$

the *gap* between the best arms and the other ones, with $\Delta = +\infty$ when all arms have the same mean. Note that the gap is not affected by ties: several arms may be best, and any of them is an equally good answer.

Definition 1. Let $\eta > 0$ and $\delta \in (0,1)$. An algorithm is *(η,δ)-PAC* if, for any instance, it returns an η-optimal arm with probability at least $1-\delta$. Its *sample complexity* is the total number of pulls it performs.

The simplest (η,δ)-PAC algorithm, described in Algorithm 1, pulls every arm the same number of times and returns an arm with the smallest empirical mean (Even-Dar et al., 2006). Its guarantee is stated below, as all other proofs, the proof is deferred to Appendix **??**.

Algorithm 1 Uniform Best-Arm Identification

Require: arms $[n]$, precision η, confidence δ
$N \leftarrow \lceil (2/\eta^2)\ln(2n/\delta)\rceil$
for $i \in [n]$ **do**
  pull arm i N times, observing $X_i^{(1)},\dots,X_i^{(N)}$
  $\hat\theta_i \leftarrow \frac{1}{N}\sum_{\ell=1}^{N} X_i^{(\ell)}$
end for
return $\hat\imath \in \operatorname{argmin}_{i\in[n]} \hat\theta_i$

Lemma 1. *Algorithm 1 is (η,δ)-PAC, with sample complexity $n\lceil(2/\eta^2)\ln(2n/\delta)\rceil$. Moreover, if $\Delta > \eta$, it returns a best arm with probability at least $1-\delta$.*

Cost independent of input-space size. Lemma 1 bounds the sample complexity solely in terms of the number of arms, precision, and confidence, independently of the reward-generation process and input-space size. This is in sharp contrast with counting. When pulls query a classifier on random completions, computing the exact error can be intractable, whereas identifying which of two candidates has lower error requires queries depending only on their gap, not on the input-space size. Thus, ranking avoids the scalability bottleneck of counting.

Errors only where they are harmless. Lemma 1 shows that when suboptimal arms are more than η from the best mean, a best arm is identified with high probability, even under ties; otherwise, any returned suboptimal arm is within η of the best. Errors therefore occur only between nearly equivalent options, a limitation inherent to the problem: distinguishing means 0.100 and 0.101 requires hundreds of thousands of samples (Kaufmann et al., 2016), precisely when the distinction matters least.

The logarithmic price of adaptivity. In our setting, Best-arm identification is applied over T adaptive rounds, where each instance depends on previous rounds.

The next lemma shows that PAC guarantees still compose under this adaptivity when rounds use fresh samples and split the confidence budget.

Lemma 2. *Consider T rounds, where the instance*

at round t may depend on the outcomes of rounds $1, \dots, t-1$, and is solved by Algorithm 1 with parameters $(\eta, \delta/T)$ using fresh samples. Then, with probability at least $1-\delta$, the following hold simultaneously for every round: the selected arm is η-optimal for the instance of that round, and it is a best arm whenever the gap of that round exceeds η.

Lemma 2 turns a guarantee on a single decision into a guarantee on an entire algorithm. Splitting the confidence as δ/T increases the number of pulls per round only by an additive term of order $\eta^{-2} \ln T$. For a greedy algorithm that makes $\mathcal{O}(d)$ decisions, controlling all of them simultaneously thus costs a factor of only $\mathcal{O}(\ln d)$.

Improvements for free. Both guarantees of Lemma 2 only rely on the fact that each round is solved by an $(\eta, \delta/T)$-PAC algorithm, since an η-optimal arm is necessarily a best arm when the gap exceeds η, so that Algorithm 1 can be replaced by any such procedure. Median Elimination (Even-Dar et al., 2006) removes the $\ln n$ factor and achieves a sample complexity of $\mathcal{O}\big((n/\eta^2)\ln(1/\delta)\big)$, which matches the lower bound of Mannor and Tsitsiklis (2004). Gap-dependent algorithms, such as successive elimination or LUCB (Kalyanakrishnan et al., 2012), stop sampling arms that are clearly suboptimal, and can substantially reduce the number of pulls when arms are well separated. Our framework thus directly benefits from any progress in best-arm identification. We use Algorithm 1 in our analysis for its simplicity, and discuss adaptive variants in Section 4.

4 BANDITEXPLAIN

We first recall greedy probabilistic explanation and its reliance on model counting (Section 4.1), discuss why approximate counting is ill-suited (Section 4.2), and then present BanditExplain and its guarantees (Sections 4.3–4.4).

4.1 Greedy Probabilistic Explanations

Given h, x, and a size limit $k \le d$, we seek a subset of at most k features minimizing the error:

$$S^* \in \operatorname*{argmin}_{S \subseteq [d],\ |S| \le k} \epsilon_{h,x}(S). \tag{4}$$

For $S \subseteq [d]$, let $\mathcal{U}(x_S)$ denote the uniform distribution over the completions of x_S, that is, over $\{z \in \{0,1\}^d : z_S = x_S\}$. By (2), the error of S is the probability that such a completion is misclassified:

$$\epsilon_{h,x}(S) = \Pr_{z \sim \mathcal{U}(x_S)} \big[\, h(z) \neq h(x) \,\big]. \tag{5}$$

Bounia and Koriche (2023) showed that the counting function $\mu_{h,x}$ is non-increasing and *supermodular*, that is, $\mu_{h,x}(A \cup \{i\}) - \mu_{h,x}(A) \le \mu_{h,x}(B \cup \{i\}) - \mu_{h,x}(B)$ for all $A \subseteq B \subseteq [d]$ and $i \notin B$. Thus, fixing a feature removes fewer misclassified completions as more features are fixed. This property motivates the greedy descent in Algorithm 2: starting from $[d]$, it repeatedly removes a feature causing the smallest increase in $\mu_{h,x}$ and returns the best subset of size at most k encountered. Ties are broken arbitrarily; an *execution* denotes any obtained run.

Algorithm 2 Greedy Descent with Exact Counting

Require: classifier h, instance x, size limit k
1: $S_d \leftarrow [d]$
2: **for** $j = d, d-1, \dots, 1$ **do**
3: pick $i_j \in \operatorname{argmin}_{i \in S_j} \mu_{h,x}(S_j \setminus \{i\})$ ▷ *exact counting*
4: $S_{j-1} \leftarrow S_j \setminus \{i_j\}$
5: **end for**
6: **return** some $S \in \operatorname{argmin}_{S' \in \{S_0, \dots, S_k\}} \epsilon_{h,x}(S')$ ▷ *exact counting*

When $\mu_{h,x}$ is computed exactly, Algorithm 2 satisfies the approximation guarantee (Bounia and Koriche, 2023): the output S_{GD} of any of its executions satisfies

$$\epsilon_{h,x}(S_{\mathrm{GD}}) \le \frac{e^p - 1}{p}\, \epsilon_{h,x}(S^*), \qquad p = \frac{c}{1-c}, \tag{6}$$

where S^* is an optimal solution of (4), and $c \in [0,1]$ is the curvature of $\mu_{h,x}$ over $2^{[d]}$ (Equation (**??**) in Appendix **??**, with $I = [d]$), which measures how far $\mu_{h,x}$ is from being modular; the bound holds whenever $c < 1$. The approximation factor $(e^p - 1)/p$ tends to 1 when c tends to 0, in which case greedy descent is optimal, and degrades gracefully as the interactions between features become stronger.

Algorithm 2 is efficient whenever $\mu_{h,x}$ is polynomial-time computable, as for decision trees. Beyond such models, its only non-trivial operation is exact counting: every decision merely compares candidate errors. During descent, candidates $S_j \setminus i$ have equal size, so comparing $\mu_{h,x}(S_j \setminus \{i\})/2^{d-j+1}$ is equivalent to comparing $\epsilon_{h,x}(S_j \setminus \{i\})$; the final step compares errors directly. Thus, every decision is an argmin over error probabilities. We focus on greedy descent rather than on the GA variant of Bounia and Koriche (2023), whose guarantee relies on a precondition that cannot be established from black-box queries (Appendix **??**).

4.2 Limits of Approximate Counting

A natural way to extend Algorithm 2 is to replace exact counts by an approximate counter such as Ap-

proxMC (Chakraborty et al., 2016), which returns, for a given S and with high probability, an estimate $\hat{\mu}_{h,x}(S)$ within a multiplicative factor $1+\gamma$ of $\mu_{h,x}(S)$. This approach faces two difficulties.

First, multiplicative guarantees hold for each estimate individually, and do not preserve the structure on which the guarantee (6) relies. In particular, the estimated function $\hat{\mu}_{h,x}$ need not be supermodular, even when every estimate satisfies its guarantee (example is given in Appendix **??**). The guarantee (6) can thus only be transferred if the approximate algorithm makes the same decisions as an exact one. The following result shows that this requires a prohibitive precision.

Proposition 1. *Let $S_1, S_2 \subseteq [d]$ be such that $\mu_{h,x}(S_1) = \mu_{h,x}(S_2) + 1 = m + 1$ with $m \geq 1$. A multiplicative guarantee of factor $1+\gamma$ on both estimates ensures that $\hat{\mu}_{h,x}(S_1) > \hat{\mu}_{h,x}(S_2)$ if and only if $\gamma < \sqrt{1+1/m} - 1$, which is of order $1/(2m)$.*

Since m can be as large as $2^{d-|S|}$, reproducing exact decisions may require exponentially small γ, while the cost of approximate counting grows polynomially with $1/\gamma$. **Second**, and more fundamentally, approximate model counters require a logical encoding of the classifier and calls to an NP oracle on this encoding, which is precisely what becomes impractical for tree ensembles and neural networks.

Best-arm identification addresses both difficulties. It only requires the ability to query the classifier as a black box. It does not attempt to reproduce the exact decisions at any cost, but explicitly accepts η-optimal decisions, with an additive precision η on error probabilities that is chosen by the user and paid for at a cost of order η^{-2}. And, by Lemma 1, it recovers the decisions of exact counting whenever suboptimal candidates are well separated from the best ones.

4.3 The Algorithm

The key observation is that each argmin of Algorithm 2 is a best-arm identification problem whose arms can be pulled at the cost of a single query to h. Consider a decision among candidate subsets $C_1, \ldots, C_n$. We associate an arm with each candidate C_i, and define a pull of this arm as follows: draw a completion $z \sim \mathcal{U}(x_{C_i})$, by setting $z_{C_i} = x_{C_i}$ and drawing the remaining features uniformly at random, and return

$$X_i = \mathbb{1}\big[\, h(z) \neq h(x) \,\big]. \tag{7}$$

By (5), X_i is a Bernoulli random variable whose mean is exactly the error of the candidate:

$$\theta_i = \mathbb{E}[X_i] = \epsilon_{h,x}(C_i). \tag{8}$$

The best arms are thus the candidates of minimal error, which are exactly the candidates that Algorithm 2 may select. Note that the reduction applies uniformly to both kinds of decisions. In the descent loop, the arms are the j candidates $S_j \setminus \{i\}$; in the final step, they are the $k+1$ subsets $S_0, \ldots, S_k$, whose sizes differ. Since arms directly estimate errors rather than counts, no normalization is needed.

BanditExplain, given in Algorithm 3, replaces each exact argmin of Algorithm 2 by a call to Algorithm 1. Since the algorithm makes $d+1$ decisions, the confidence budget is split as $\delta/(d+1)$, following Lemma 2.

Algorithm 3 BanditExplain

Require: black-box classifier h, instance x, size limit k, precision η, confidence δ

$\delta' \leftarrow \delta/(d+1)$, $\quad S_d \leftarrow [d]$
for $j = d, d-1, \ldots, 1$ **do**
  $i_j \leftarrow$ Algorithm 1 on arms $\{S_j \setminus \{i\} : i \in S_j\}$ with (η, δ') ▷ *bandit*
  $S_{j-1} \leftarrow S_j \setminus \{i_j\}$
end for
return Algorithm 1 on arms $\{S_0, \ldots, S_k\}$ with (η, δ') ▷ *bandit*

Comparing Algos 2 and 3 shows that BanditExplain preserves the structure of greedy descent, and only changes how decisions are made: by sampling and querying the classifier, rather than by counting. Each pull requires drawing $d - |C_i|$ random bits and evaluating h once, so that BanditExplain applies to any classifier that can be queried, without any encoding.

4.4 Theoretical Guarantees

We now state the guarantees of BanditExplain. We say that a decision, that is, an argmin over candidates $C_1, \ldots, C_n$, is *η-separated* if its gap (3), with arm means given by (8), exceeds η. Equivalently, every candidate whose error is not minimal has an error exceeding the minimal one by more than η. In particular, a decision in which several candidates are tied for the minimal error may be η-separated.

Theorem 1. *Let $\eta > 0$ and $\delta \in (0,1)$. With probability at least $1-\delta$, the following hold simultaneously:*

(i) every removal made by Algorithm 3 is η-optimal, that is, $\epsilon_{h,x}(S_{j-1}) \leq \min_{i \in S_j} \epsilon_{h,x}(S_j \setminus \{i\}) + \eta$ for every $j \in [d]$;

(ii) the returned subset S_{BE} satisfies $|S_{\mathrm{BE}}| \leq k$ and $\epsilon_{h,x}(S_{\mathrm{BE}}) \leq \min_{0 \leq \ell \leq k} \epsilon_{h,x}(S_\ell) + \eta$;

(iii) if every decision faced by Algorithm 3 is η-separated, then S_{BE} is the output of an execution of Algorithm 2, and therefore $\epsilon_{h,x}(S_{\mathrm{BE}}) \leq \frac{e^p-1}{p}\,\epsilon_{h,x}(S^*)$ whenever $c < 1$.

Moreover, Algorithm 3 performs at most $\mathcal{O}\big(d^2\eta^{-2}\ln(d/\delta)\big)$ queries to h.

Guarantees and positioning. Theorem 1 offers two complementary guarantees. By *(i)–(ii)*, every decision of BanditExplain is η-near-optimal *i.e.,* it selects a candidate whose error exceeds the best one. Under η-separation,(iii) guarantees that BanditExplain follows a path that exact greedy descent could have taken, and hence inherits its approximation guarantee (Bounia and Koriche, 2023). Separation does not require a unique best candidate: ties are harmless, as greedy descent may break them arbitrarily. These guarantees hold for any classifier, including models for which exact error computation is intractable. Unlike locally-minimal explanations (Izza et al., 2024), which only ensure that no single feature can be removed from the returned subset, our guarantees relate each decision to a provably near-optimal greedy procedure. These guarantees hold for any classifier, including those for which computing $\epsilon_{h,x}$ is intractable. Compared with Anchors (Ribeiro et al., 2018), which uses best-arm identification to find a rule meeting a prescribed precision threshold, BanditExplain targets the error-minimizing candidate and inherits the approximation guarantee of greedy descent under separation.

Query complexity. BanditExplain requires $\mathcal{O}(d^2\eta^{-2}\ln(d/\delta))$ classifier queries, polynomial in d, $1/\eta$, and $\ln(1/\delta)$, and independent of model size. Thus, its query cost does not grow with the size of a tree ensemble or the depth of a neural network, beyond the cost of a single forward evaluation. Queries for a given decision are independent and can be evaluated in a single batch.

Adaptive variants. Items (i)–(ii)–(iii) of Theorem 1 hold for any (η, δ')-PAC best-arm identification algorithm. Algorithm 1 can therefore be replaced by Median Elimination or LUCB. Gap-dependent methods are particularly attractive: when many candidate features are irrelevant, they can be discarded after few queries, concentrating the sampling budget on the candidates that are difficult to distinguish.

5 EXPERIMENTAL EVALUATION

To assess the practical value of our theoretical results, we benchmarked BanditExplain against an exact SAT-based method (Arenas et al., 2022), exact greedy descent (Bounia and Koriche, 2023), and two scalable baselines, Anchors (Ribeiro et al., 2018) and locally-minimal explanations (Izza et al., 2024). Our evaluation has two main objectives: *(a)* Measure the accuracy of BanditExplain against a ground truth: on decision trees, the error $\epsilon_{h,x}$ of any explanation can be computed exactly in polynomial time, allowing a direct comparison with exact methods. *(b)* Demonstrate its scalability on complex models, namely tree ensembles and neural networks, for which exact methods are no longer tractable. These *2* studies are complemented by an illustration on image classification and by a sensitivity analysis with respect to η and δ. All methods were implemented in Python 3.14, and all experiments were run on a laptop with an Intel Core i7-1165G7 CPU at 2.80 GHz and 16 GB of RAM. Further details and complete results are available in Appendix **??** and in `https://tinyurl.com/BanditExplain`.

5.1 Experimental Setup

Experiments were conducted on 16 benchmarks from the OpenML, UCI and Kaggle repositories, covering tabular and image data. All benchmarks were mapped to a Boolean domain $\{0,1\}^d$, with each feature indicating whether an input satisfies a condition. Numerical attributes were discretized into three quantile-based bins using Scikit-learn's `KBinsDiscretizer` (Pedregosa et al., 2011), while categorical attributes were unchanged. For images, features correspond to 2×2 pixel blocks. Each benchmark defines an explanation task (h, x, k). Decision trees were trained on 15 datasets ($8 \leq d \leq 784$); random forests, XGBoost, and one-hidden-layer MLPs (64 ReLU units) were trained on Pima, Tic-Tac-Toe, and Heart ($d \leq 14$), where errors remain exactly enumerable. For images, we used a CNN with two convolutional layers (16 and 32 filters) and a $32 - \mathit{unit}$ dense layer on MNIST and STL-10. Neural networks used Adam; all models were trained on 70% and evaluated on 30% of each dataset. We set $k \in [3, 7]$ by dataset, following human cognitive constraints (Miller, 1956), and varied it from 2 to 15 for sensitivity analysis.

Default hyperparameters for BanditExplain were set to $\eta = 0.1$ and $\delta = 0.05$. The SAT-based method proceeds by binary search on the error, with precision 10^{-3} and a time limit of 600 seconds per instance. On five of the 15 datasets, this search was interrupted and returned explanations that are not optimal (orange rows of Table 1). Since Anchors does not take a size limit as input, the comparison is made at equal size: the baseline is run first, the size s of its explanation is recorded, and BanditExplain is then run with $k = s$; the same protocol is used for both baselines on complex models. For each benchmark, all methods were evaluated on 150 test instances, and we report the

Dataset	Acc	#Inst.	d	k	SAT	GD	BE	LmPAXp	BE = GD (%)	BE ($k = s$)	Anchors
Wine	59.83	1143	11	7	0.1093 ± 0.1728	0.2044 ± 0.1171	0.2052 ± 0.1188	0.2098 ± 0.1244	90.67	0.1137 ± 0.1889	0.2292 ± 0.3146
Adult	72.67	6061	12	7	$0.1584 \pm 0.1444^{\dagger}$	0.1430 ± 0.0898	0.1446 ± 0.0918	0.1513 ± 0.0975	84.67	0.2815 ± 0.1447	0.4661 ± 0.2807
Loan	74.05	614	12	7	0.0128 ± 0.0427	0.1374 ± 0.1678	0.1376 ± 0.1689	0.1414 ± 0.1751	70.67	0.1429 ± 0.1791	0.2778 ± 0.3570
Heart	99.03	1025	14	7	0.0968 ± 0.0989	0.1117 ± 0.0693	0.1125 ± 0.0697	0.1191 ± 0.0767	99	0.2829 ± 0.0724	0.3768 ± 0.1951
Bank	73.33	11162	16	7	–	0.1245 ± 0.1076	0.1253 ± 0.1089	0.1291 ± 0.1116	76.67	0.1761 ± 0.1756	0.3567 ± 0.3048
Compas	90.56	7214	18	5	0.0005 ± 0.0109	0.0166 ± 0.0370	0.0172 ± 0.0371	0.0200 ± 0.0394	98.00	0.1924 ± 0.0804	0.3847 ± 0.0603
Twonorm	84.33	7400	21	7	$0.2100 \pm 0.1321^{\dagger}$	0.1805 ± 0.0946	0.1823 ± 0.0953	0.1851 ± 0.0978	79.33	0.3999 ± 0.0732	0.4889 ± 0.0758
Mushrooms	99.75	8124	23	3	$0.1992 \pm 0.1977^{\dagger}$	0.1833 ± 0.1957	0.1833 ± 0.1960	0.1848 ± 0.1987	100	0.1895 ± 0.1647	0.2255 ± 0.1903
Breast	91.81	569	33	4	$0.2436 \pm 0.1775^{\dagger}$	0.1872 ± 0.1038	0.1878 ± 0.1043	0.1925 ± 0.1058	99.33	0.2162 ± 0.0520	0.4312 ± 0.1401
Ionosphere	83.96	351	35	4	$0.2020 \pm 0.1635^{\dagger}$	0.1790 ± 0.1415	0.1789 ± 0.1420	0.1770 ± 0.1428	100	0.1880 ± 0.1084	0.4085 ± 0.2748
Kr-vs-Kp	98.85	3196	37	4	0.0192 ± 0.0346	0.0300 ± 0.0541	0.0308 ± 0.0557	0.0313 ± 0.0570	89.33	0.0504 ± 0.0532	0.1197 ± 0.1555
Spambase	89.43	4601	57	7	–	0.1691 ± 0.1058	0.1730 ± 0.1083	0.1790 ± 0.1145	63.33	–	–
Mnist49	95.70	13782	784	7	–	0.3491 ± 0.0987	0.3581 ± 0.1017	0.3654 ± 0.1045	51.33	–	–
Mean difference w.r.t. GD					–	0	+0.00160	+0.00538	–	–	–
Number of datasets won					–	–	12/13	1/13	–	–	–

Table 1: Explanation errors (mean ± std). BE = GD: percentage of instances with the same error. BE ($k = s$): BE run with the size s of the explanation of Anchors. Orange (†): SAT search timed out. Blue (–): no result.

mean and the standard deviation of the exact error. The sensitivity analysis, on four datasets, is reported in Section 5.5.

5.2 Ground-Truth Evaluation

Table 1 reports the results on (13/15) decision trees; the full results are in Appendix **??**. BE has a lower error than LmPAXp on 12/13 datasets. At equal size, it also beats Anchors on all 11 datasets where Anchors could be run, often by a wide margin: on Adult, BE reaches 0.28 against 0.47. BE also stays close to exact GD. Its error is 0.0016 above GD on average, less than a third of the gap of LmPAXp, and the two methods return exactly the same error on 85% of the instances. On Mnist49, the hardest case with $d = 784$, the difference is still below 0.01.

The SAT-based solver returns the optimum when it terminates, which happens on five datasets. On Heart, Compas, and Kr-vs-Kp, BE is within 0.02 of this optimum. On Wine and Loan the gap is larger, but it is the gap of greedy descent itself, which BE matches. On five other datasets (orange rows), the SAT search hit the time limit and its explanations are less accurate than those of BE (0.244 against 0.188 on Breast). On the last three (blue rows), which include the two of largest dimension, it returned nothing. BE returned an explanation on every instance.

Figure 1 plots the error gap to GD against the size limit k. On Adult and on Bank, the curve of BE is never above that of LmPAXp. BE stays within 0.4 percentage points of GD, while LmPAXp reaches 1.2 points on Adult and 1.3 on Bank. The difference shows mostly for small explanations ($k \leq 6$), where each retained feature matters more. For large k, both gaps go to zero because the error itself vanishes. As for cost, BE takes between 0.13 s and 41 s per instance on average depending on the dataset (median 6.1 s). LmPAXp takes 0.93 s on average and GD 0.008 s, but the polynomial-time counting used by GD is specific to decision trees. The explanations returned by BE always contain at most k features.

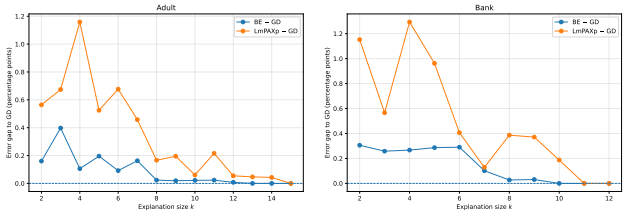


Figure 1: Error gap to GD, in % points, as a function of the size limit k, on Adult and Bank. Lower is better.

5.3 Evaluation on Complex Models

We now consider models for which $\mu_{h,x}$ cannot be derived from the model structure: random forests, XGBoost, and multi-layer perceptrons. BE only needs query access to them. On low-dimensional datasets, the error of an explanation S can still be computed exactly by querying the model on all $2^{d-|S|}$ completions of x_S, so the methods are compared without estimation noise. The optimum, however, is out of reach as a general reference: the problem is $\mathrm{NP}^{\mathrm{PP}}$-complete (Wäldchen et al., 2021), and exhaustive search takes up to 3^d evaluations. We therefore compare the scalable methods with each other, at equal size.

Table 2 gives the results on Pima and Heart, which follow the same trends as on decision trees. BE has a lower error than Anchors on all six configurations, by 33% on average (from 0.106 to 0.071). Its explanations are never larger on average, and it is also faster. BE is more accurate than LmPAXp on all six configurations too, but by a smaller margin (7% on average). LmPAXp is two to four times faster, yet the advantage of BE holds across the three model families. In every configuration, BE runs in under two seconds per

				Comparison with Anchors			Comparison with LmPAXp		
Dataset	Model	d	Acc	Anchors	BE ($k = s$)	Change	LmPAXp	BE ($k = s$)	Change
Pima	NN	8	79.65	0.1035 ± 0.1183	$\mathbf{0.0686} \pm 0.0827$	-33.7%	0.0963 ± 0.0898	$\mathbf{0.0929} \pm 0.0881$	-3.5%
	RF		86.58	0.0242 ± 0.0849	$\mathbf{0.0154} \pm 0.0455$	-36.4%	0.0162 ± 0.0292	$\mathbf{0.0148} \pm 0.0273$	-8.6%
	XGB		87.88	0.0381 ± 0.1016	$\mathbf{0.0283} \pm 0.0698$	-25.7%	0.0185 ± 0.0260	$\mathbf{0.0169} \pm 0.0256$	-8.6%
Heart	NN	14	80.19	0.2490 ± 0.1572	$\mathbf{0.1599} \pm 0.1107$	-35.8%	0.0657 ± 0.0502	$\mathbf{0.0584} \pm 0.0483$	-11.1%
	RF		98.05	0.0894 ± 0.1016	$\mathbf{0.0568} \pm 0.0628$	-36.5%	0.0170 ± 0.0284	$\mathbf{0.0158} \pm 0.0272$	-7.1%
	XGB		98.05	0.1344 ± 0.0995	$\mathbf{0.0960} \pm 0.0712$	-28.6%	0.0351 ± 0.0313	$\mathbf{0.0336} \pm 0.0334$	-4.3%
Mean error				0.1064	**0.0708**	-33.4%	0.0415	**0.0387**	-6.6%
Configurations won				0/6	**6/6**		0/6	**6/6**	

Table 2: Exact explanation errors (mean $\pm$ std) on complex models, over 150 test instances. BE ($k = s$): BE run with the size s of the explanation of the baseline. Change: relative change in error of BE with respect to the baseline. Lower error in bold. NN: multi-layer perceptron, RF: random forest, XGB: XGBoost.

instance and returns explanations with fewer than five features on average. Complete results are given in Appendix **??**.

5.4 Illustration on Image Classification

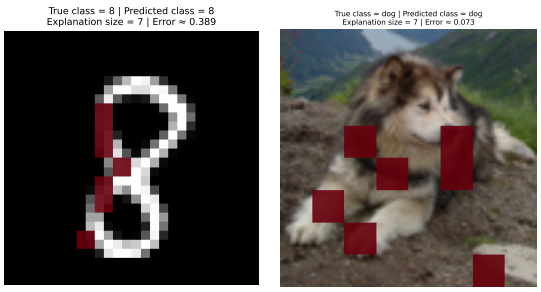


Figure 2: Explanations of size $k = 7$ returned by BE (in red) for the same CNN on MNIST, and on STL-10.

Finally, we illustrate BE on image classification, using the same convolutional network in two very different regimes. On MNIST it reaches 99.65% test accuracy; on STL-10 (cats versus dogs), without any tuning, it only reaches 66.31%. We ran BE with $k = 7$, and Figure 2 shows the selected blocks in red. On MNIST (3 versus 8), all seven blocks fall on the left strokes of the digit, those that close its two loops. This is what separates an 8 from a 3, whose loops are open on the left. On STL-10, most of the selected blocks are on the animal, on the chest just below the muzzle and on the front legs, and not on the background. In both cases, seven blocks give an explanation that agrees with human intuition, even for a classifier of modest accuracy. The estimated error is 0.39 on MNIST, with 7 of the 196 blocks, and 0.07 on STL-10. Further examples are given in Appendix **??**.

5.5 Sensitivity to η and δ

Last, we study how the precision η and the confidence δ affect BE, on the decision trees of Adult, Bank, Loan, and Tic-Tac-Toe. We varied both parameters together, from $(0.10, 0.05)$ to $(0.60, 0.30)$, on the same 150 instances and with k fixed. In Table 3, the running time follows the $\eta^{-2} \ln(d/\delta)$ trend predicted by Theorem 1. It drops by more than a factor of four when η is doubled, and by a factor of 46 to 55 between the two extremes. Over all 20 configurations, the observed speed-up is within 15% of the prediction. The error degrades much more slowly: it increases by at most 0.008 when η goes from 0.1 to 0.2, and by less than 0.07 at $\eta = 0.6$. Increasing η is therefore a simple way to trade accuracy for speed. Full results, and the same study on neural networks, are in Appendix **??**.

	Speed-up (obs. / pred.)		Error		
Dataset	$\eta = 0.2$	$\eta = 0.6$	0.1	0.2	0.6
Adult	**4.4** / 4.6	**53.3** / 53.5	0.144	0.152	0.201
Bank	**4.3** / 4.5	**54.3** / 52.2	0.129	0.132	0.172
Loan	**4.4** / 4.6	**45.9** / 53.5	0.151	0.158	0.216
Tic-Tac-Toe	**4.2** / 4.6	**55.4** / 55.0	0.065	0.068	0.087

Table 3: Sensitivity of BE to η (with $\delta = \eta/2$). Speed-up with respect to $\eta = 0.1$: observed (bold), and predicted (gray) by the bound $\eta^{-2} \ln(d/\delta)$ of Theorem 1.

6 CONCLUSION

We introduced BanditExplain, a method that computes probabilistic explanations with query access to the model only, by casting each step of greedy descent as a best-arm identification problem. With $O(d^2 \eta^{-2} \ln(d/\delta))$ queries, it guarantees with probability at least $1 - \delta$ that every removal is η-optimal, and it coincides with exact greedy descent whenever the candidates are η-separated. Experimentally, it matches exact greedy descent on decision trees, improves on Anchors and locally-minimal explanations on tree ensembles and neural networks, and remains applicable where exact methods fail, up to $d = 784$. BanditExplain thus combines query-only scalability with per-decision PAC guarantees, and with the guarantee of exact greedy descent whenever decisions are separated, two requirements that existing approaches meet only separately. Two directions are particularly

promising: extending BanditExplain beyond uniform distributions, where sampling remains possible even when counting does not, and applying the same best-arm reduction to other explanation problems.

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